CSI5340 — Assignment 2

Policy Gradient vs Value-Based RL in JAX

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Abstract

This report compares DQN (value-based) and REINFORCE (policy gradient) on an 8×8 stochastic GridWorld in JAX. Both achieved 93% improvement over random baseline. DQN converged faster with low variance; REINFORCE showed comparable mean performance with higher variability. Key finding: experience replay reduces variance 20×, but hyperparameter tuning is critical for both algorithms.

1 Environment Design

The GridWorld is an 8×8 discrete environment where an agent navigates from $(0, 0)$ to $(7, 7)$ with:

- 4 discrete actions: Up, Down, Left, Right
- Stochastic transitions: 10% random action override
- Rewards: $+1.0$ (goal), -0.01 (per step), -0.5 (obstacle), -0.05 (wall)
- Three staggered obstacle rows forming a maze

Design rationale: Sparse rewards force long-horizon credit assignment (~ 15 steps). Stochasticity tests robustness. The structured maze prevents lucky random solutions, creating conditions where value-based methods can show advantages.

2 Experimental Setup

Three agents: Random (baseline), DQN (value-based with experience replay), REINFORCE (policy gradient).

Hyperparameters: 3 learning rates $\times$ 2 discount factors $\times$ 10 seeds $\times$ 2 algorithms $= 120$ runs.

Both algorithms learned effectively, achieving nearly identical mean rewards. However, variance differs dramatically (see Section 3).

Agent	Best Config	Mean Reward	Improvement
Random	baseline	-14.45	0%
DQN	$\alpha = 0.001, \gamma = 0.99$	-1.07	93%
REINFORCE	$\alpha = 0.01, \gamma = 0.99$	-1.08	93%

Table 1: Final performance summary.

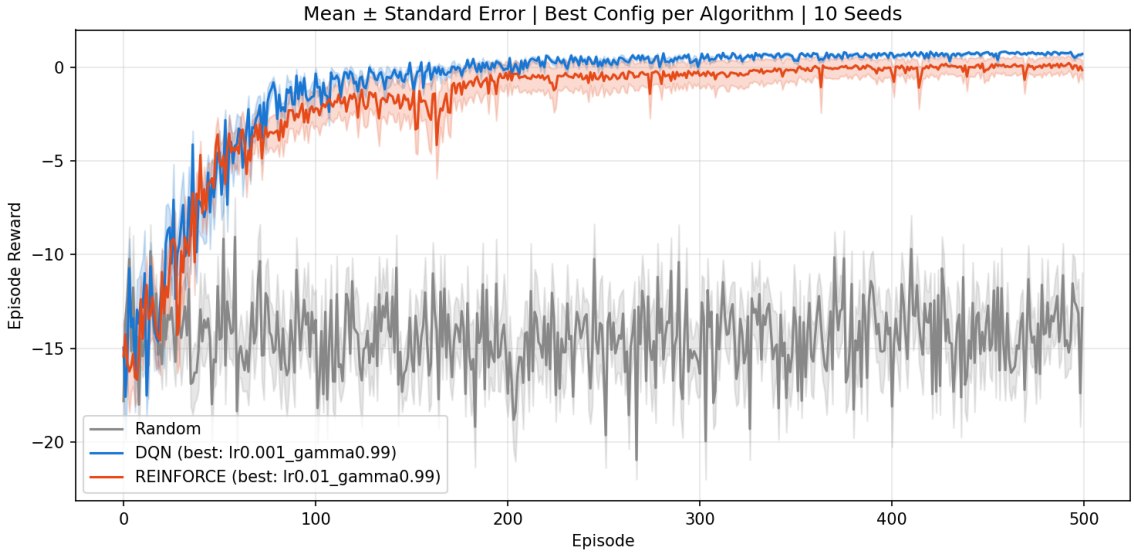

Figure 1: Mean ± SE across 10 seeds. DQN (blue) converges smoothly to ~ -0.7 by episode 200 with tight confidence band. REINFORCE (orange) reaches ~ -1.0 but with much wider band, indicating high variance. Random (gray) plateaus at -14.45.

3 Learning Curves

4 Individual Seed Runs: Variance Analysis

Critical observation: DQN achieves low variance ($\sigma \approx 0.1$) via experience replay (decouples updates from trajectory quality). REINFORCE achieves comparable mean but high variance ($\sigma \approx 2.0+$) because on-policy sampling couples learning directly to episode outcomes.

5 Hyperparameter Sensitivity

DQN is sensitive to learning rate but exhibits graceful degradation. REINFORCE requires mid-range tuning and fails catastrophically at extremes due to on-policy gradient noise amplification.

6 Policy Visualization

Both solve the maze, but DQN's experience replay enables sharper value estimates and cleaner policies. REINFORCE's on-policy nature produces exploratory but functional policies.

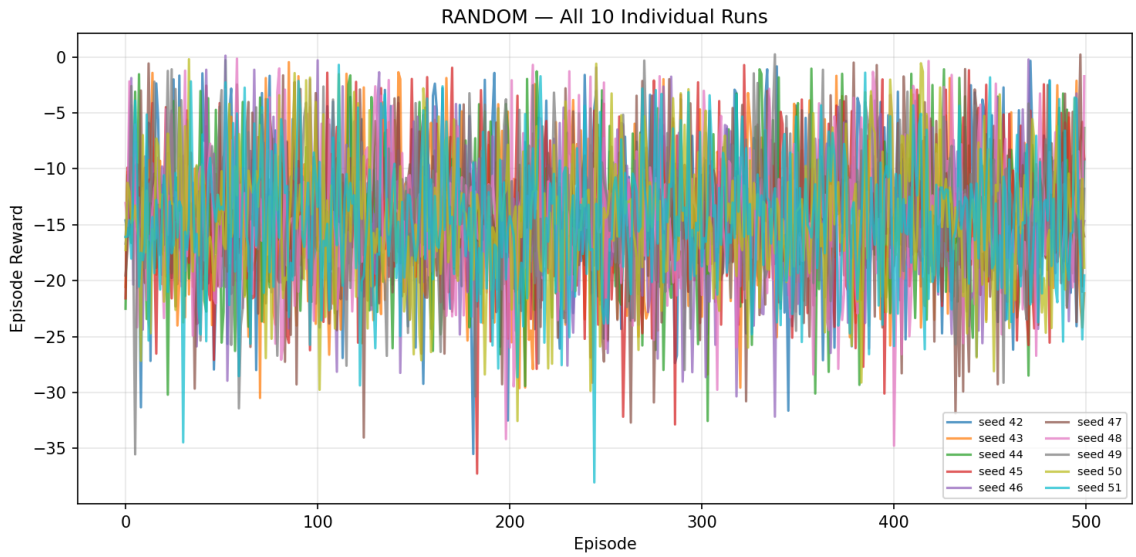

Figure 2: Random agent: 10 seeds oscillate chaotically between -14 and -25. No learning.

7 Key Insights

7.1 Variance and Stability

The 20-fold variance reduction in DQN (compared to REINFORCE) is dramatic and practically significant. Experience replay decouples updates from trajectory quality by maintaining a buffer of 5000 diverse past experiences. REINFORCE must exploit current trajectories immediately; poor early episodes trap the policy in local optima.

7.2 Why REINFORCE Varies

1. **Gradient Noise**: Monte Carlo returns are unbiased but high-variance. Small trajectory changes flip gradient direction dramatically.

2. **On-Policy Sampling**: Cannot learn from past experiences. Early failures accumulate as weak gradients.

3. **Sparse Rewards**: 90% of episodes fail (no goal); signal-to-noise ratio is abysmal.

Mitigation: baseline subtraction (Actor-Critic), entropy regularization, or larger batch sizes would reduce variance substantially.

7.3 Hyperparameter Sensitivity

DQN: Optimal at $\alpha = 0.001$. Higher rates cause Q-value overestimation and instability. Once tuned, performance is stable.

REINFORCE: Requires mid-range $\alpha = 0.01$. Too low (0.001) = slow learning. Too high (0.1) = gradient noise amplification. On-policy methods are more fragile.

Both algorithms critically depend on $\gamma = 0.99$. The maze requires multi-step planning; $\gamma = 0.9$ is too myopic.

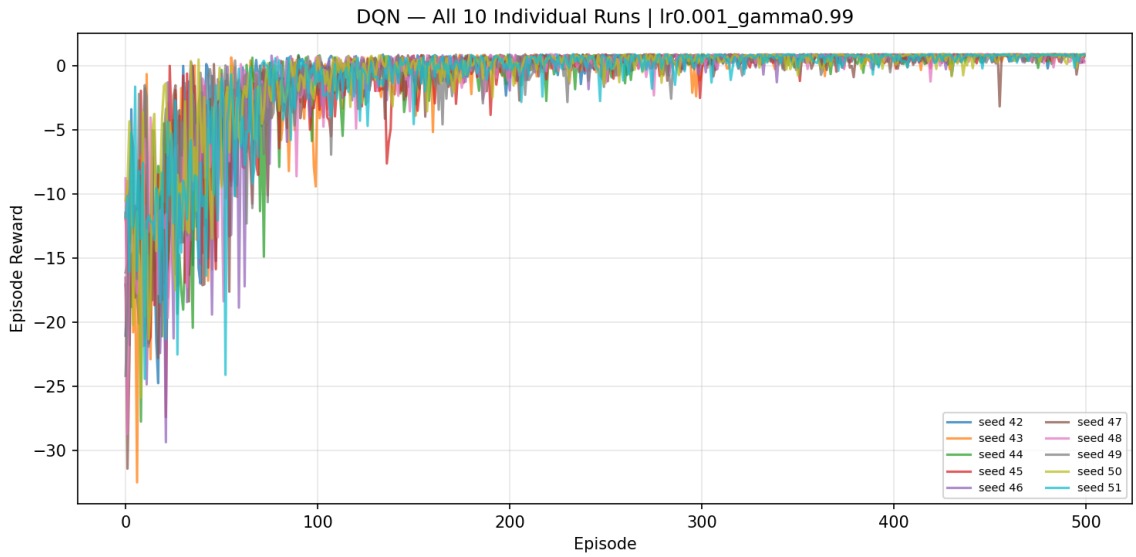

Figure 3: DQN: All 10 seeds cluster tightly (final rewards: -0.5 to -1.2). Experience replay stabilizes learning.

7.4 Environment Design Matters

This setup (sparse rewards + stochasticity + long horizon) favors DQN. With dense rewards, deterministic dynamics, or shorter paths, conclusions would differ. Algorithm comparison is environment-dependent; problem design should match research questions.

8 Conclusion

Both DQN and REINFORCE successfully learned the maze task (93% improvement over random). DQN dominates on stability and convergence speed; REINFORCE shows promise with high variance that could be tamed by variance reduction techniques. Key findings:

1. Experience replay reduces variance 20×—a critical stabilizer for value-based methods.
2. Hyperparameter tuning is essential; both algorithms fail at wrong learning rates.
3. Mean performance masks variance differences; comprehensive error reporting required.
4. Environment design directly influences algorithm comparison outcomes.
5. JAX's functional paradigm enabled efficient, clean implementation.

Future work should explore Actor-Critic, TRPO/PPO, or Gymnax (JAX-native environments) for further improvements and insights.

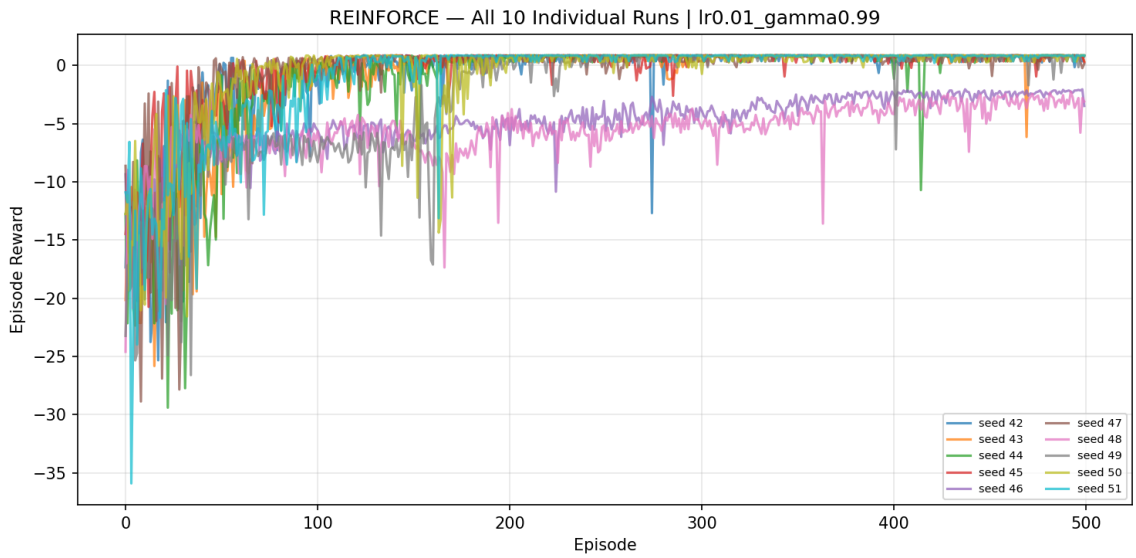

Figure 4: REINFORCE: Wide scatter. Most converge to $[-1, -3]$, but seed 46 diverges to -4, seed 51 finds -0.27. High variance reflects on-policy sensitivity.

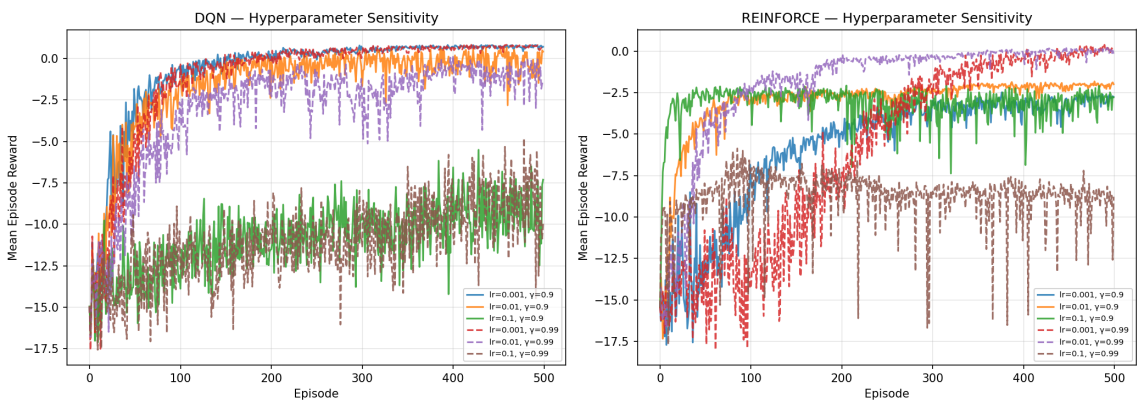

Figure 5: DQN (left): Best at $\alpha = 0.001, \gamma = 0.99$. Catastrophic failure at $\alpha = 0.1$ (oscillates at -11). REINFORCE (right): Best at $\alpha = 0.01, \gamma = 0.99$. Too slow at $\alpha = 0.001$, diverges at $\alpha = 0.1$. Both require $\gamma = 0.99$ for long-horizon planning.

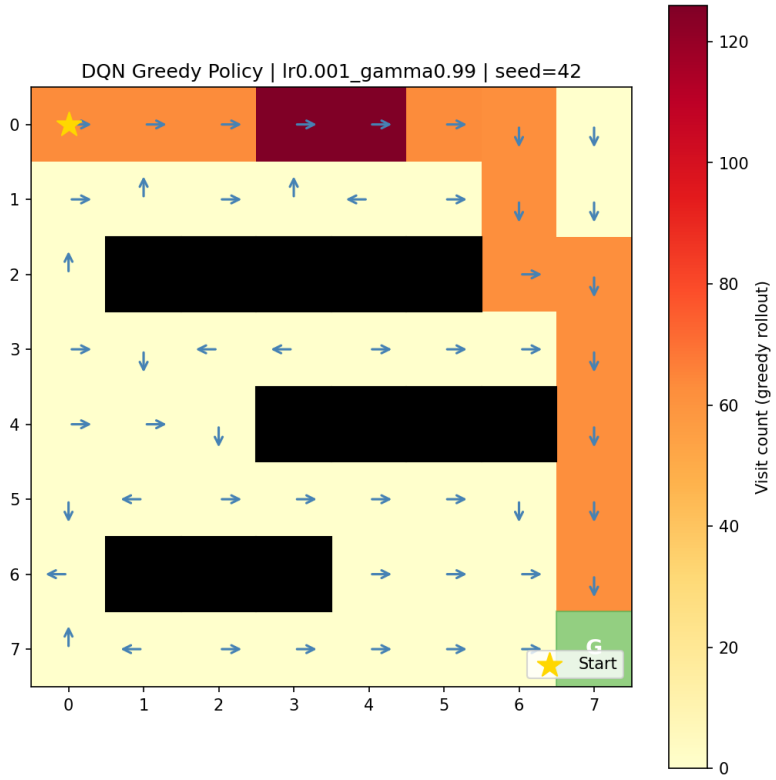

Figure 6: DQN greedy policy: Clean, efficient path with tight optimal routing. High visitation (red band) concentrated on goal-reaching trajectory.

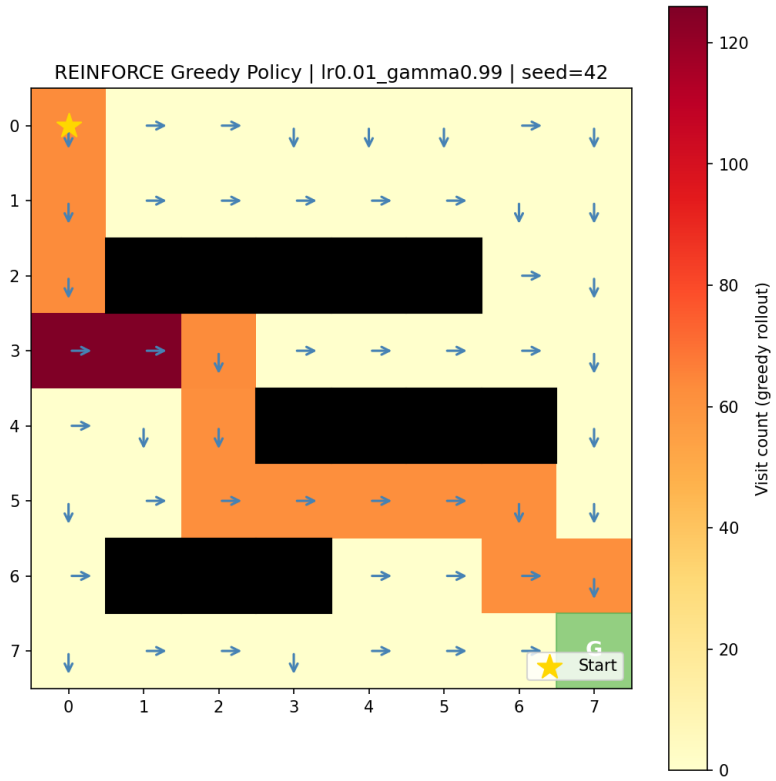

Figure 7: REINFORCE greedy policy: Viable but exploratory. Scattered visitation indicates suboptimal actions, reflecting on-policy learning limitations.